

**TECHNOLOGICAL UNIVERSITY DUBLIN
TALLAGHT CAMPUS**

Bachelor of Science (Honours)

Pharmaceutical Science

Full Time

Semester Eight : May 2022

Manufacturing Technology

Internal Examiners

Dr Adrienne Fleming
Dr Eugene Hickey

External Examiners

Dr Ken Kinsella

Day
Date May 2022
Time

Instructions to Candidates

Section A: A Fleming: Answer any two out of the three questions.

Section B: E Hickey: Answer any two out of the three questions.

Section A **Answer any two questions**

Question 1 **Answer all parts** **(100 marks)**

- (a) ***'The pharmaceutical industry produces between 25 and 100 kg or more of waste per kilo of API manufactured'.***

Discuss this statement with relation to potential methods to reduce the extent of waste produced.

(30 marks)

- (b) Determine the rate of heat generation (in Watts, assuming Q calculated in J s^{-1}) from the reaction of 22 moles of reactant with HCl in water at 15°C in a 25 m^3 vessel, where ΔH for the reaction was found to be $-25.4 \text{ kJ mol}^{-1}$.

The activation energy was calculated as being 134 J mol^{-1} while the reaction rate constant was determined as being $3.2 \times 10^{-4} \text{ s}^{-1}$. [$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$].

(30 marks)

- (c) The maximum temperature of a synthetic reaction (MTSR) is the sum of the process temperature and the adiabatic rise in temperature.

Given that you are required to carry out a reaction at 40°C in acetonitrile as a solvent in large scale production, determine the MTSR for the reaction, i.e. the temperature above which decomposition can occur (in $^\circ\text{C}$), given that the total mass of all reagents in the vessel was 356 kg, and the number of moles of limiting reagent was 47.6 moles, with $\Delta H = -387 \text{ kJ mol}^{-1}$ and the specific heat = $1.5 \text{ kJ kg}^{-1}\text{K}^{-1}$.

(40 marks)

Question 2 **Answer all parts** **(100 marks)**

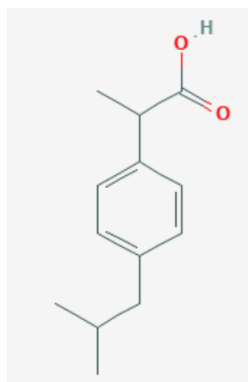
- (a) An open straight-blade turbine of 0.88 m in diameter is used to stir reaction mixtures in a 25 m³ batch reactor with baffles. The vessel diameter is 1.6 m. The mixture to be stirred is composed of 10% NaOH solution with phenyl acetate at reflux. Addition of the acetate takes place into the agitator region.

Given that the agitator speed has been fixed at 180 r.p.m., determine the mixing time (t_{95}) in seconds.

Given that $Po = 0.92$; $\rho = 1490 \text{ kg/m}^3$; $\mu = 0.015 \text{ Ns/m}^2$

(30 marks)

- (b) Assess the following compound as a reagent in large scale production from a safety perspective? Explain why, using your knowledge of oxygen balance. **(20 marks)**



- (c) A 20 m³ batch reactor is fitted with a flat blade turbine agitator of diameter 1.2 m and is set to operate at 90 r/min. The reactor is to be used to react 7.58 moles of reactant A with reactant B at room temperature. The homogeneous reaction mixture has a density of 1956 kg m⁻³ and a viscosity of 1.3 Ns m⁻² with a calculated rate constant of 3.12 s⁻¹. In this process reactant A is to be added in a controlled fashion down the walls of the vessel. Using time constant calculations determine whether product selectivity is determined by micromixing or reaction kinetics. [$P_0 = 0.88$].

(20 marks)

- (d) *“You are a member of a team that has been tasked with looking at the design of the chemical reactor at full scale production”.*

Discuss the factors that you see to be important in the design process.

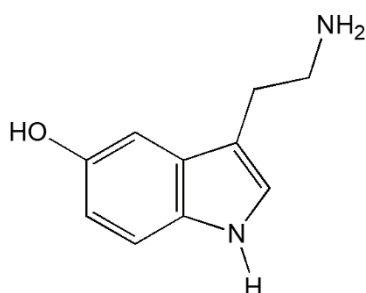
(30 marks)

Question 3 **Answer all parts** **(100 marks)**

(a) "Polymorphism is a phenomenon that affects the performance of Drugs"

Discuss the importance of the control of crystallisation the API to ensure the correct polymorphic form is produced. **(30 marks)**

(b) Discuss the potential structural modifications to the following structures that could be varied to enhance drug potency and activity in the body.



(20 marks)

(c) A new plant is being designed for the production of the API Naproxen.

Given that the installation cost of a 10m³ reactor in 2011 has been calculated to be €122,500, predict the cost in 2016, given the following CEI cost index data:

CEI (2011) = 420 and CEI (2016) = 525

(20 marks)

(d) 'Process Analytical Technologies are systems that enhance process understanding and assist in identifying and controlling critical points in a process'.

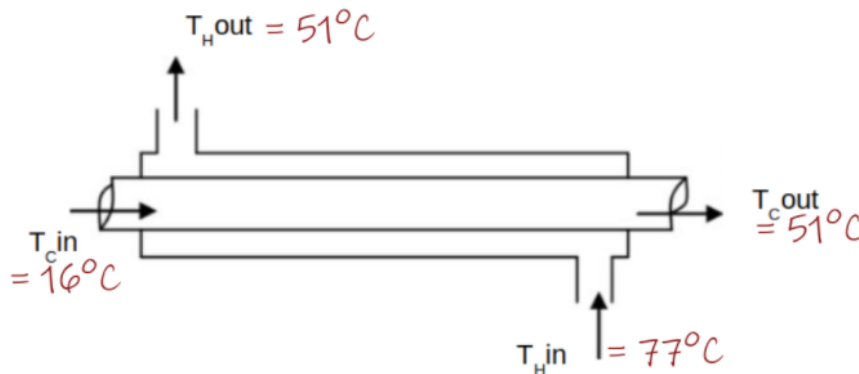
In your own words, discuss this statement. Use examples to support your answer.

(30 marks)

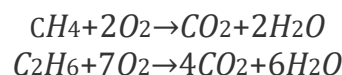
Section B Answer any two questions

Question 4 Answer all parts **(100 marks)**

- a) In a shell and tube heat exchanger, what factors dictate which fluid should pass through the shell and which through the tubes? (10 marks)
- b) What is meant by the **Log Mean Temperature Difference (LMTD)** and what role does it play in heat exchanger analysis? (10 marks)
- c) Water (heat capacity = $4190 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1}$) flowing at a rate of 2 kg/s enters a counter flow heat exchanger at 16°C and leaves at 51°C . It is heated by an oil stream (heat capacity = $1890 \text{ J kg}^{-1} \text{ K}^{-1}$) entering at 77°C , leaving at 51°C .



- i) Sketch a profile of the temperature across this heat exchanger for these two streams. (10 marks)
- ii) Show that the LMTD for this process is 30.28°C . (10 marks)
- iii) Show that the rate of heat gain by the water stream is $2.933 \times 10^5 \text{ W}$. (10 marks)
- iv) Calculate the flow rate of the oil stream. (15 marks)
- v) Calculate the UA value of the heat exchanger. (15 marks)
- d) To ensure complete combustion, 30% excess air is supplied to a furnace burning 200 kmol/hr of natural gas. The v/v composition of natural gas is 95% ethane and 5% ethane. Air contains 21% oxygen by mole. Calculate the number of kmol/hr of oxygen required. The reactions are:



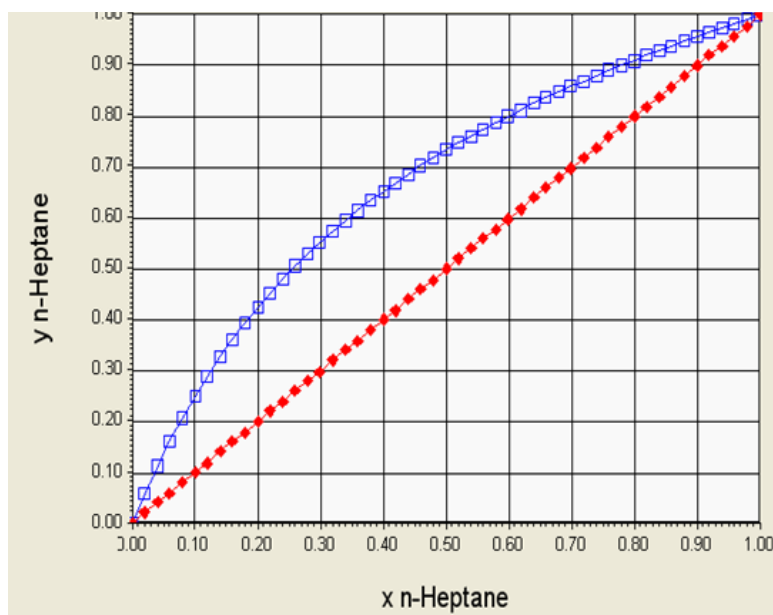
(20 marks)

Question 5

Answer all parts

(100 marks)

- a) A feed steam with a total flow of 400kmol/hr is passed in to a distillation column. The composition of the feed stream by mole is 40% methanol and 60% water. The top product from the column is 92% methanol. The bottom product is 10% methanol. Calculate the total flow rates in kmol/hr of the top and bottom product.
(35 marks)
- b) The diagram below shows the VLE plot for n-heptane and ethyl benzene at 101.3kPa.
- Explain what is plotted on the x and y axes of a VLE diagram.
(10 marks)
 - What is the composition of the vapour phase when the mole fraction of n-Heptane in the liquid phase is 0.4?
(10 marks)
 - What is the composition of the liquid phase when the mole fraction of ethyl benzene in the vapour phase is 0.3?
(10 marks)
 - Calculate the relative volatility of n-heptane and ethyl benzene at 101.3kPa when the mole fraction of n-heptane in the liquid phase is 0.35.
(20 marks)



- c) Three problems that may arise with inappropriately designed distillation columns are:
- flooding
 - weeping
 - entrainment

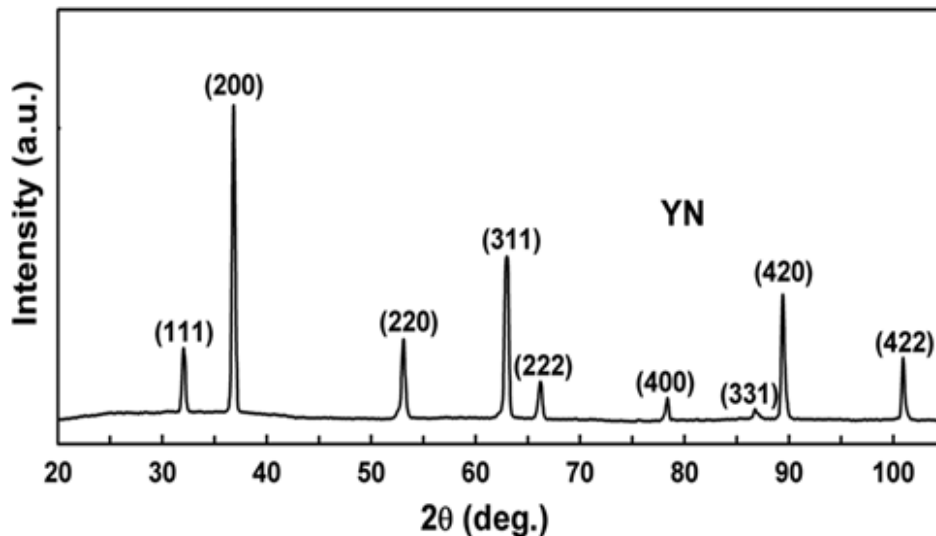
Explain what is meant by each of these terms and how each effects distillation column performance
(15 marks)

Question 6

Answer all parts

(100 marks)

- a) Discuss in one paragraph the role of the 14 Bravais lattices in crystal structure under the following headings:
- The reason why there are only seven possible shapes for the Bravais Lattices.
 - The four possible flavours of Bravais lattice.
 - Why there are only 14 (and not $7 \times 4 = 28$) Bravais Lattices.
- (20 marks)
- b) The diagram below shows a powder diffraction pattern use Cu K α radiation from the cubic crystal, Yttrium Nitride (YN).



- i) Find the 2θ position of all the peaks (20 marks)
- ii) Find the inter-planar spacing for all the peaks (20 marks)
- iii) Use one of the inter-planar spacings (you can choose whichever you want) to find the lattice constant (20 marks)
- iv) Is YN primitive, BCC, or FCC? (20 marks)

LIST OF FORMULAE

MT42

$$\text{Cost in Year A} = \text{Cost in Year B} \times \frac{\text{Cost Index in Year A}}{\text{Cost Index in Year B}}$$

$$C_2 = C_1 (S_2/S_1)^n$$

$$\text{Production Cost} = \text{annual production cost} / \text{annual production rate}$$

$$Q_r = V(-\Delta H_r)k_0 C_A \exp(-E_A/RT) \quad Q_c = UA(T-T_a)$$

$$\text{MTSR} = T_p + \Delta T_{ad} \quad \Delta T_{ad} = \Delta H.n / m.C_p$$

$$\text{TMR}_{ad} = C_p R T^2 / q_0 E_A$$

$$\text{Oxygen balance} = \frac{-[1600(2a + b/2 - c)]}{\text{Mol. Wgt.}}$$

Batch Conversion Time

$$t = -1 / k \ln(A_0 - A_t)$$

$$\text{Reaction Time constant } \tau_R = 1/kC_{A0}$$

$$\text{Re} = \frac{\rho v D}{\mu}$$

$$\text{Re} = \frac{\rho N D^2}{\mu}$$

$$P = P_0 N^3 D_A^5 \rho$$

$$\varepsilon_{\text{mean}} = P / V\rho \quad (\text{Reminder } \varepsilon_{\text{local}})$$

$$\text{Re}_c = 6370 / P_0^{1/3}$$

$$P_0^{1/3} \text{Re Fo} = 5.2$$

$$\text{Fo} = \frac{\mu t_{95}}{\rho T^2}$$

The lifetime of an Eddy is τ_K :

$$\tau_K = 12[\nu / \varepsilon]^{1/2}$$

where $\nu = \mu / \rho$ (kinematic viscosity)

$$\text{Engulfment Rate } E = \ln 2 / \tau_K$$

$$\tau_E = 1/E$$

$$q = mc\Delta T$$

$$q = UA \, LMTD$$

$$LMTD = \frac{\Delta T_L - \Delta T_R}{\log_e(\Delta T_L / \Delta T_R)}$$

$$\text{Molar mass CO}_2 = 44\text{g/mol}$$

$$\text{Molar Mass Air} = 29\text{g/mol}$$

$$n\lambda = 2d \sin \theta$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

$$\lambda_{Cu \, K\alpha} = 0.154051\text{nm}$$

$$\alpha_{AB} = \frac{y_A / x_A}{y_B / x_B}$$

$$q = \text{liquid fraction}$$

$$\text{slope} = \frac{-q}{1-q}$$

$$q\text{-line goes between } (x_F, y_F) \text{ and } (0, x_F/(1-q))$$

$$\text{rectifying line between } (x_D, y_D) \text{ and } (0, x_D/(R+1))$$

$$F = D + B$$

$$F x_F = Dx_D + Bx_B$$